

Department of Statistics and Computer Science
University of Peradeniya
CS 202 - Data Structures Practical (2020/21)
Lab Sheet 04
STACKS

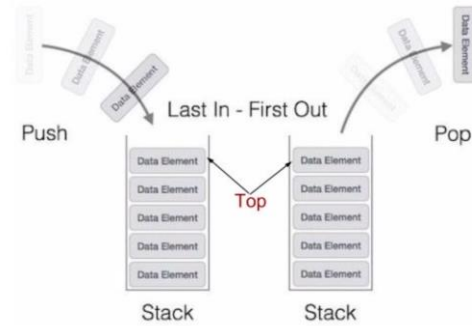
Purpose

- Understanding the implementation of Stack data structure using linked Lists.
- Get an idea of applications of stacks.

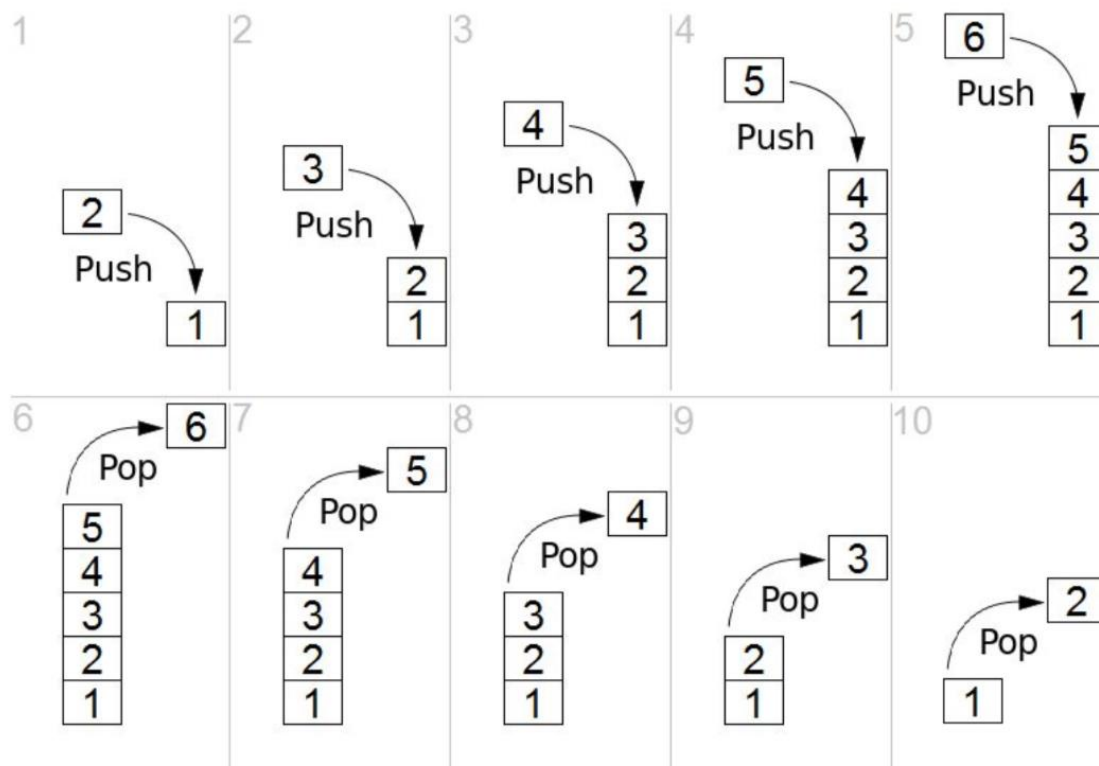
Theory

- A stack is an Abstract Data Type (ADT) which can be implemented using either an array or a linked list.
- Based on the LIFO (Last In First Out) algorithm.
- Access (Both insertion and Retrieval) is restricted to the beginning of the list which is called the **top** of the stack.
- Differences from a normal linked list,
 - Stack has only one reference in its stack structure called **top** instead of the two links (**head** and **tail**) in the linked list.
 - Insertion of elements (declared as **push** function) is only done from the top/beginning of the stack (similar to insert from front of linked lists).
 - Retrieval of elements (declared as **pop** function) is also done from the top/ beginning of the stack (similar to delete from front of a linked list).
- The basic stack operations consist of,
 - **isEmpty** (Optional) - check whether the stack is empty (returns *true* if empty, *false* otherwise.)
 - **push** - insert nodes from the top
 - **pop** - retrieve value of node at the top (removes the element/node from the stack)
 - **peek** - retrieve value of the top node without removing the node from the stack
 - **display** - print all the nodes in the stack into system output. (starting from the top)

- Visual Implementation of a stack



- Usage Example



Question 01

- Implement the stack data structure by implementing the functions mentioned above.
- Create a main function and do the following,
 - Declare and initialize a stack that can be used to store some integer values.
 - Store the following values in the stack,
 - 1, 5, 6, 6, 7, 8, 10, 38
 - Retrieve a value from the stack
 - Insert 58, 43
 - Retrieve another value from the stack and insert 100 and 378 after the retrieval.
 - Print the contents of the stack.

Question 02

1. A palindrome is a phrase which would be the same when read beginning to end or end to beginning.

Eg: MADAM

ABBA

Write a program that can be used to check whether phrase given by a user is a palindrome or not using only stacks as data structures.

(Hint: you may use more than one stack in order to accomplish this).

2. An infix expression is an expression of the form **a op b**. When an operator is in-between every pair of operands. A postfix expression is an expression of the form a b op. When an operator(op) is followed for every pair of operands. For example, the infix expression **a+b** can be expressed as **ab+** in postfix expression. Though postfix expressions are easily and efficiently evaluated by computers, they can be difficult for humans to read. Therefore we would like to convert the postfix expressions to infix for the purpose of reading and editing.

Write a program to convert the given postfix expression into an infix expression using stacks.

Ex:

- | | |
|----------------------|----------------------|
| I. Input : ab*c+ | Output : ((a*b)+c) |
| II. Input : abc++ | Output : (a+(b+c)) |
| III. Input : ab+cd+* | Output : (a+b)*(c+d) |